# FIELD TERMINATION GUIDE



Pinouts, polarity and color codes for AV and lighting installs.

**HOUSE RULES**

**Black is never hot.**

**Feed-Through RJ45 connectors are not allowed**

Audio XLR: **pin 2 = +** · DMX: **pin 3 = Data +**

Label both ends before you strip.

**02** XLR-3 balanced audio
**03** XLR connector faces
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**Device-specific terminations & updates**

illuminated-integration.github.io/field-terminations

**illuminated-integration.com**
**717 996 4596 · Harrisburg PA**

# XLR-3 BALANCED AUDIO

## XLR-3 PINOUT (AES14)

| PIN | SIGNAL | CONDUCTOR |
|---|---|---|
| 1 | **Shield / Gnd** | drain / braid |
| 2 | **Hot (+)** | colored / light |
| 3 | **Cold (–)** | black / dark |

**1-SHIELD · 2-HOT · 3-COLD**

- Read the pin numbers molded beside each solder cup; don't rely on face orientation from memory.
- Sleeve the drain before it reaches pin 1 so it can't touch pin 2.

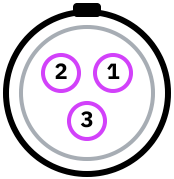
## PAIR COLORS · HOUSE STANDARD

| PAIR | PIN 2 (+) | PIN 3 (–) |
|---|---|---|
| Red / Black | Red | Black |
| Red / White | Red | White |
| White / Black | White | Black |

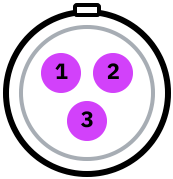
- Shield to pin 1 at both ends. Leave the shell lug open unless the spec says otherwise.
- Match what's already on the job before mixing standards.
- Never use mic cable for DMX (see card 06).

# XLR CONNECTOR FACES

## XLR-3 · FRONT VIEW, LATCH UP




**FEMALE**
socket face




**MALE**
pin face

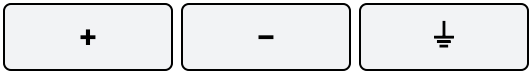
| PIN | AUDIO | DMX |
|---|---|---|
| 1 | Shield | Common |
| 2 | **Hot +** | **Data –** |
| 3 | **Cold –** | **Data +** |

**Same connector, opposite polarity.**
Pin 2 is **+** for audio and **–** for DMX.
Never re-use an audio cable for DMX.

- Solder side is the mirror image of the front view.
- The numbers molded on the insert are the final word.
- XLR-5 pinout: card 06.

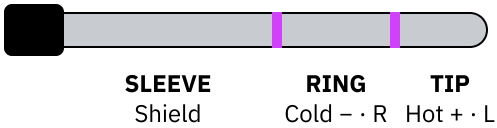
# EUROBLOCK & TRS / TS

## EUROBLOCK / PHOENIX, 3-POSITION

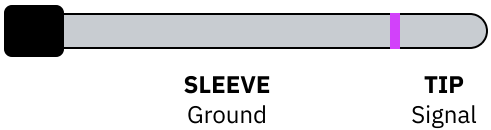



- **Order varies by manufacturer.** Follow the silkscreen, not habit.
- Strip to the device's spec (typ. 5–7 mm). No bare copper past the clamp.
- **Don't tin wire** for screw terminals; solder creeps and loosens. Ferrules if they fit.
- Tug-test every wire. Heat-shrink any lifted shield.

## ¼" / 3.5 MM TRS · BALANCED OR STEREO




## TS · UNBALANCED




**Balanced out → unbalanced in:** + to tip; – and shield to sleeve at the unbalanced end. Some outputs must not have – grounded; check the manual.

# SPEAKERS: SPEAKON & 70 V

## SPEAKON NL4 / NL2

| CONTACT | FULL-RANGE | BI-AMP* |
|---|---|---|
| 1+ |  + | LF + |
| 1– |  – | LF – |
| 2+ | — | HF + |
| 2– | — | HF – |

*Common, not universal. NL2 = 1± only. Check the speaker manual.

## SPEAKER WIRE POLARITY

- Red / Black jacketed: **Red = +**, Black = –. **NEVER** use Ferrules!

## 70 V / 100 V LINES

- Line – to transformer COM; line + to the wattage tap.
- Verify tap setting from drawing specification.
- Never land a 70 V line on the 8 Ω tap. Don't mix 70 V and 100 V taps.
- Polarity check: tester, or a 9 V pop on a passive woofer (+ to +, cone moves out).

# DMX PINOUTS: 3 & 5-PIN

## DMX512 XLR (E1.11)

| PIN | SIGNAL |
|---|---|
| 1 | **Common / Shield** |
| 2 | **Data 1 –** |
| 3 | **Data 1 +** |
| 4 | Data 2 – (spare) |
| 5 | Data 2 + (spare) |

**1-SHIELD · 2-DATA– · 3-DATA+**

- **Daisy-chain only.** No Y-cables; use an opto-splitter.
- **Terminate** the last device: 120 Ω across pins 2 & 3.
- 32 unit loads per segment, max.
- DMX needs 120 Ω cable. **Mic cable will cause errors.**
- 3-pin is common but non-standard. 3↔5 adapters run pin-for-pin 1-2-3.

**PAIR 1 · Black & Red**

Always the primary DMX

- Red → Data +
- Black → Data –
- Drain → Common

**PAIR 2 · Black & White**

5-pin spare, else fold back

- White → Data 2 +
- Black → Data 2 –
- Drain → Common

| WIRE | SIGNAL | XLR |
|---|---|---|
| Red ① | **Data +** | 3 |
| Black ① | **Data –** | 2 |
| Drain ① | **Common** | 1 |
| White ② | Data 2 + | 5 |
| Black ② | Data 2 – | 4 |
| Drain ② | **Common** | 1 |

**TWO BLACK WIRES.** Keep each black with its own foil and twist. The black twisted with **red** is Data –. Pair 2 black only ever goes to pin 4.

Block = push-in IN/THRU terminal (card 08). Pair 2 on pins 4/5 is house standard.

# 9729 → PUSH-IN TERMINAL KIT

**Kit:** HS1 ×2 drain sleeve · HS2 ×1 jacket · HS3 ×2 foil · HS4 ×1 spare-pair cap · 2 cable ties. **Tools:** stripper, flush cutters, heat gun, 3 mm flat blade.

1. Label the cable (DMX #). Strip **190 mm** of jacket.
2. Separate the pairs. Cut foils back to **10 mm** from the jacket.
3. **HS1** over both drains, leaving 10 mm bare. Shrink.
4. **HS3** over each foil stub. Shrink.
5. **HS2** over the jacket end and stubs. Shrink.
6. Fold the spare pair back, cap with **HS4**, tie it to the jacket. Tie the working pair 30 mm from its ends.
7. Strip **10 mm**: Red → 3 / 6, Black → 2 / 5, Drain → 1 / 4.
8. Push a **3 mm** blade into the release (A), insert the wire (B), pull the blade, tug-test.
9. IN only: 1-2-3. IN/THRU: incoming on 1-2-3, outgoing on 4-5-6.

# DMX OVER CAT CABLE

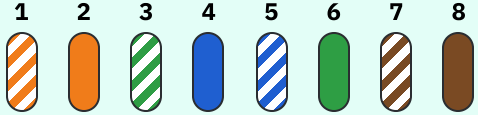
**RJ45 ↔ XLR · ANSI E1.27-2, T568B**

| RJ45 | WIRE | SIGNAL | XLR |
|---|---|---|---|
| 1 |  Wh / Org | **Data 1 +** | 3 |
| 2 |  Orange | **Data 1 –** | 2 |
| 3 |  Wh / Grn | Data 2 + | 5 |
| 6 |  Green | Data 2 – | 4 |
| 7 |  Wh / Brn | **Common** | 1 |
| 8 |  Brown | **Common** | 1 |
| 4, 5 |  Blue pair | not used | — |

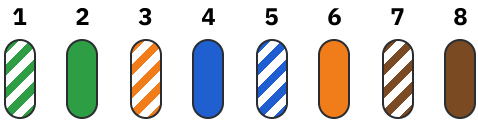
- Cat5e or better, UTP or STP.
- **Not every vendor follows E1.27-2.** Some pixel / show-control gear and some IDC kits use their own RJ45 map. Check the manual.
- DMX on Cat cable is **not Ethernet**. Label it DMX and never patch it into a switch.

# ETHERNET: T568A / T568B

**T568B · house default**




**T568A**




Plug view: contacts toward you, latch away, **pin 1 on the left**.

- Same standard both ends = straight-through.
- Untwist **≤ 13 mm (½")**. Jacket goes under the crimp strain relief.
- Pass-through plugs: trim flush, no copper whiskers.
- Shielded: bond the drain to the plug or jack shield.
- 100 m channel max (90 m permanent + 10 m patch).
- Wiremap-test every run. Certify all runs carrying Dante, HDBaseT, or PoE.

# LC FIBER & STRAND COLORS

## POLARITY




**A-to-B duplex:** TX always lands on RX.

- No link? Flip the duplex clip / uniboot polarity.
- **Inspect → clean → inspect** before every mate.
- **Never look into a port or fiber end.** Use a VFL.
- No kinks or crushed ties; use hook-and-loop.

## CONNECTOR BODY COLOR (TIA)

| | |
|---|---|
| Singlemode UPC | |
| **Singlemode APC** (8°) | |
| OM1 62.5 µm | |
| OM2 50 µm | |
| OM3 / OM4 (OM4 may be violet) | |
| OM5 | |

**Never mate APC to UPC.** Green only goes to green.

## STRAND & TUBE ORDER · TIA-598

13–24 repeat with a black tracer

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Blue | Orange | Green | Brown | Slate | White | Red | Black | Yellow | Violet | Rose | Aqua |

# COAX: RG6, RG8 & RG8X

## AT A GLANCE

| | RG6 | RG8 | RG8X |
|---|---|---|---|
| **Impedance** | **75 Ω** | **50 Ω** | **50 Ω** |
| Size | ~0.27" OD | ~0.40" OD | ~0.24" OD |
| Connectors | F, BNC 75 | N, BNC 50, UHF | BNC 50, UHF |

**Never mix 50 Ω and 75 Ω.** BNC comes in both: 75 Ω for SDI video, 50 Ω for wireless antennas. Check the marking.

- Use the compression connector made for that exact cable (RG6 and RG6 quad differ).
- No braid strands near the center pin.

## RG6 75 Ω

- **SDI video** (HD / 3G / 12G) on precision RG6
- CATV, off-air antenna, satellite

## RG8 50 Ω

- **Wireless mic & IEM antenna runs** (remote paddles)
- Cellular / Wi-Fi antenna feeds (LMR-400 class)

## RG8X 50 Ω

- Short antenna jumpers where RG8 is too stiff to dress
- Higher loss than RG8: keep runs short

Long antenna runs: lower loss beats thinner cable. **VERIFY** from manufacturer impedance required of cable.

# CONTROL: RS-232 / 485 / GPIO

## RS-232 · DB9 (DTE / COMPUTER SIDE)

| DB9 | SIGNAL | LANDS ON |
|---|---|---|
| 2 | RX | far-end TX |
| 3 | TX | far-end RX |
| 5 | Ground | Ground |

- Phoenix Tx / Rx / G: **cross Tx↔Rx**, ground to ground.
- Most AV control uses 3 wires. No handshake unless the protocol doc asks for it.
- No response? Swap Tx/Rx first, then check baud / parity / stop bits.

## RS-485 / 422

- Data on one twisted pair, plus common. **A/B naming isn't consistent** between vendors; if it's dead, swap A/B.

## RELAYS & GPIO

- Relay: COM + **NO** (open at rest) or **NC** (closed at rest).
- Confirm GPIO voltage and pull-up before landing a contact closure.

# CLOSEOUT CHECKLIST

## BEFORE YOU LEAVE THE RACK

- [ ] Both ends labeled to the drawing's cable ID
- [ ] Continuity + polarity tested (audio / speaker)
- [ ] DMX: last device terminated
- [ ] Cat: wiremap pass, certified where required
- [ ] Fiber: inspected, cleaned, light-level checked
- [ ] Coax: correct impedance connectors, signal verified
- [ ] Strain relief, service loop, no copper showing
- [ ] as-builts updated

**Device-specific pinouts** can be found here. Missing something? Submit a request!

## JOB NOTES